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“Economic impact of pedestrianisation in historic urban centre, the Valdemoro case – study (Spain)”

Julián Sastre ^a, Ana Sastre ^b, Ana Maria Gamo ^c, Tomás Gaztelu ^d

^a, * *Vice president of Spanish Infrastructures and services Forum, Avda Kansas City, 28 Sevilla 41007. Spain*

^b, *Khairros, Sebastian Recasens 36, Sevilla 41007 Spain*

^c *Valdemoro Town Council, C/ Gaspar Bravo de Sobremonte s/n Valdemoro (Madrid) 28340 Spain*

^d *Advisor, C/ Manantial, 130 Oficina-124 Edificio CEI Bahía de Cádiz, El Puerto de Santa María, (Cádiz) 11100 Spain*

Abstract

Building activities in urban centres involving pedestrianisation are never bereft of controversy. In this paper we shall look at the methodology applied by the Authors in the Valdemoro case (a town of around 60,000 inhabitants located within the metropolitan area of Madrid) in order to evaluate the economic impact on the local area of the pedestrianisation programme which has been planned and is currently being carried out in stages as part of the Sustainable Mobility Plan to which the town adheres.

- Firstly, a benchmarking was undertaken in which different experiences of pedestrianisation were assessed, and through the application of a Delphi method results analysis, management and expert groups were able to decipher the responses.
- Secondly, a specific market study was undertaken in the Valdemoro area through contacts with local trading associations, people from the business world to quantify the specific impact which this type of construction might have. Contacts were made with commercial representatives in the town: trade associations and guilds, business centres, the local Chamber of Commerce, The Central Traders' Association, and which are others as yet unspecified. Taking into account what has been mentioned beforehand and the benchmarking results, an assessment is made using a model which allows us to:
 - Perform recommendations on which businesses to develop.
 - Evaluate the impact on the area as a whole.
 - Detect the most widely affected businesses.
 - To fix the conditions to guarantee success from the economic viewpoint of the pedestrianisation programme.

* Corresponding author: **Julián Sastre González**. Independent Consultant
Tel.: 0034 670502757
juliansastre@juliansastre.com

The application and results of this innovative methodology are shown below.

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1. General thoughts on pedestrianisation

The analysis of experiences demonstrates that improvements can be made in the following aspects:

1. **Environmental Improvement.** An environmental improvement can be achieved, in the short term, by reducing noise and atmospheric pollution, and by decreasing the number of accidents, etc.
2. **Tourism.** Mostly in the more important cities, with historic centres, there are enhanced to attract more visitors (both from home and abroad) which bring these areas to life and contribute economically to improving the immediate environment and local economy of the city. Historic buildings are often given drastic makeovers and new lighting so that they become even more popular with tourists.
3. **Identity.** Greater identification and representativeness of the local residents and inhabitants further afield will be achieved through the creation of a space which provides comfort, tranquillity etc, in such a way that this environment becomes identifiable as their own, as belonging to the everyday life of the city and forms part of the day to day customs and habits of its inhabitants.

In general, for those actions which improve the pedestrian network, the focal points of the city become revitalised, enhancing shopping and leisure facilities and reducing work and study related elements.

4. **Aesthetics.** Normally the implantation of pedestrianised areas come with an improvement of the urban landscape through the re-urbanisation of the entire area, the renovation of urban buildings and decorative or artistic elements, the integration of significant constructions and the recovery or setting-up of elements of Urban Heritage etc, all of these enhance and facilitate the use of these areas.
5. **Cultural.** Cultural events such as street theatre, buskers, artistic performances etc which serve to bring life to the area, in which shoppers, tourists, chance visitors and those just out for a walk intermingle.
6. **Social.** There is a marked increase in the social nature of the pedestrianised area, as this becomes the fulcrum of the redeveloped area. Pedestrianised areas constitute an element which tempers and civilizes spaces which have all too often become aggressive and overcrowded.
7. **The Economy .**There is also an increase in commercial activity which brings with it a consequential rise in economic improvement, the opening of new business ventures, an increase in demand, the creation of new jobs and positions, increased tax revenue, a rise in property values etc. .

The necessary conditions for this boost to the economy to take effect are:

- Proper planning of the entire process
- Supply of public parking facilities in the areas around the intended development to cater for increased demand.
- An adequate supply of public transport facilities which links this development to the rest of the city.
- A design structure which strengthens potential aspects such as accessibility, a pleasant environment and lighting.

2. A Selection Of Flagship Pedestrianisation Programmes In Spain

We shall briefly look at a selection of examples which are worthy of special mention. Of the most recent examples is that of Fuencarral Street in Madrid which beforehand suffered as a major traffic black-spot served by narrow pavements on which the myriad of pedestrians often lacked space due to the excessive demand provided by the local shopping area.



Fig 1: New image of Fuencarral Street

In the historic city of Seville, besides the famous Santa Cruz neighbourhood, an ever-popular with tourists, two of the cities most important avenues have been pedestrianised:

- Asuncion Street, a popular shopping area and the gateway to Seville's world renowned fair.

Nuevos itinerarios de circulación de corto y largo recorrido

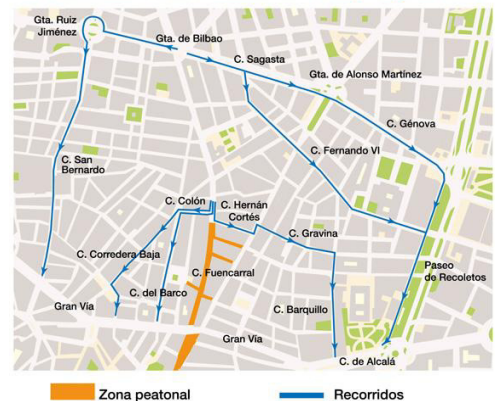


Fig 2: Pedestrian itineraries



Fig 3: View of Constitution Avenue before and after the pedestrianisation programme



Fig 4: View of Asuncion Street before and after the pedestrianisation programme

- Constitution Avenue, the main access road to the city centre which was transformed into a pedestrianised avenue where cyclists, pedestrians and the ultra-modern tram users co-habited.



Fig 5: Seville Metrocentre

Finally, one has to mention the city of San Sebastian where the pedestrianisation of the Old Town, more extensive than is usually the case in Spain. Littered with an impressive shopping area and a wide range of typical taverns and other leisure elements of interest for tourists. The city centre has been connected by a pedestrianised walkway which joins the Cathedral, as can be observed in the picture below, providing first-rate services.



Fig 6: San Sebastián

3. Economic effects as based on experiences

Many business proprietors believe that having parking spaces outside their premises is an essential element for their trade. Yet, experience shows that in many cities the conversion of these areas to pedestrianisation which allows for access solely by public transport as well as loading and unloading has actually increased their businesses' income.

The graph below shows the increase (expressed in percentages) of commercial activity following the pedestrianisation of areas in different cities.

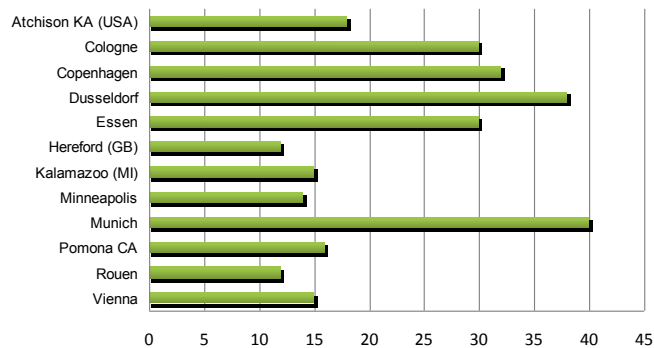


Fig 7: Practical examples of increases in commercial activity following pedestrianisation "Better Mobility in Urban Areas". UITP.

4. Methodological aspects

In this section we shall outline the methodology followed by the authors as a result of their different experiences:

1. **Definition of the study's scope:** Limits are established for the study's scope as well as its range of influence for this type of development.
2. **Analysis of Historical Precedents:** Marking the steps and aims needing to be taken as well as Mobility Goals in the town of Valdemoro.
3. **Socio-economic framework study:** In order to be able to understand the functioning and features of the study's scope, diverse socio-economic variables will be analysed, supported by data obtained from different sources.
4. **Benchmarking:** similar experiences which have already concluded will be analysed so that the successful elements of these can be adapted to this new project, thus evaluating its potential with regards to the study's scope. In the same way, interviews have been held with specialists in mobility and town-planning, as well as representatives of local groups and official bodies based in the area to evaluate the general feeling towards this type of project.
5. **Field-work:** In order to complete the compiled information and be able to obtain direct results regarding the current situation, a data-compiling campaign has been developed with the aim of creating an inventory of the local businesses which could be affected by a drop in commercial activity, and their posterior grouping into different pre-defined categories. we well as taking a specific inventory on each business through the creation and recording of a sheet with the current trades located in the area where it is planned a great commercial development
6. **Supply analysis:** A zonal and route analysis will be performed, whose aim is to detect the main axes and traffic flows, and to improve or enhance these depending on needs.
7. **Estimation of the economic repercussions of the aforementioned pedestrianisation.** Once the current situation is made known and following the analysis of developments in similar areas, an economic model will be produced which displays the economic repercussion in the short, medium and long term, quantifying the variations in wealth which are produced directly or indirectly on the commercial fabric of the environment under study.

			
- INVENTARIO DE COMERCIOS EN CALLES ADYACENTES A LAS VÍAS PEATONALIZABLES -			
Nombre de la Calle: Calle de San Antón			
CATEGORÍAS			
MODA Y CALZADO			
COMPLEMENTOS			
HOGAR		2	
BAZAR			
ALIMENTACIÓN Y BEBIDAS			
PAPELERÍA Y REGALOS		1	
BARES Y RESTAURANTES		1	
BANCOS, SEGUROS E INMOBILIARIAS			
OTROS			
ESPECIFICAR			



Fig 8: Inventory index card – Specimen[†]

[†] Inventory index card – Specimen. List of businesses in streets adjacent to pedestrianised areas. Street Name: San Antón: Type of Business: Footwear and Fashion, Complements. Household Goods, Bric-a-brac, Perishable goods, Stationers and Gift shops, Hostelry, Financial Services, Others.

8. As has already been mentioned, meetings and interviews have been held with experts on mobility and town-planning, as well as the heads of local trade associations and business groups in the Valdemoro area, with the aim of evaluating the proposals which are put forward by these groups, with particular attention being paid to the provisional evaluation made to this project.

We shall proceed now to a summary of the most relevant findings:

- Regarding the design and management of the project scope, citizen participation should be on hand in order to achieve the highest level of consensus possible.
- Careful attention must be paid to design and future treatment of spaces.
- One of the priorities for this project will be to achieve an area of co-existence in which residents, traders, workers who are employed in this area, visitors, tourists etc can share the space in such a way that the benefits are not enjoyed by some at the expense of others, rather that improvements bring with them a higher standard of living for all.
- A key issue for success in these areas, with regards to access to the pedestrianised area, is that the management of public parking facilities be carried out to the letter.
- In the same way, it is necessary for there to be an adequate supply of public transport facilities which allows for citizens to be able to reach the new area from any part of the city.
- Spaces must be established in which loading and unloading will take place, and even, should this be necessary restrict access at certain times.
- Municipal public services for waste collection, cleaning, security, maintenance and emergency services must be coordinated in tandem.
- Peripheral problems which may occur on the outskirts of this development must be dealt with, especially traffic, parking, Taxi-stands, loading and unloading areas, bicycle lanes etc.

5. The Valdemoro Case – Study

Valdemoro is a residential satellite town in the Madrid municipal area with a population of 60,000. The old town is where most of the administrative buildings are located though also complemented by important commercial outlets, found in the Estrella Street. The most important historical monuments located in the city are not mapped in tourists' guide books, nor are they perceived as having enhanced the city's worth as a cultural destination.



Fig 9: Examples of local heritage



Fig 10: Valdemoro. Main shopping district

In 2007 a Sustainable Urban Mobility Plan (PMUS) was undertaken in which a pedestrianisation policy was defined for important thoroughfares with the aim of reducing traffic congestion. In order to pacify the initial hostility to the plan manifested by the local traders, a study was carried out which showed the economic effects using the previously mentioned methodology.

A specific model was produced which gave the results shown below: Estimated jobs created (contracted)

Table 1. Specific model

SCENARIO		VALUE IN EUROS							
		Residential Surface Revaluation	Commercial Surface Revaluation	Variation in value due to usage change	Overall increase in property value	Income generated for the Public Administration	Market wealth generated	Estimated jobs created (self- employed)	Estimated jobs created (contracted))
ZONE A	MAIN THOROUGHFARE	20,088,487	4,878,613	12,358,206	37,325,306				
	SECONDARY THOROUGHFARE	3,991,136	2,055,029	4,573,743	10,619,908	4,959,360	12,512,455	146	73
ZONE B	ROAD NETWORK	6,521,067	2,509,812	-1,018,569	8,012,310				
TOTAL		30,600,691	9,443,454	15,913,380	55,957,525	4,959,360	12,512,455	146	73

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